

Signal2Scan

Can routine ICU data predict head-CT yield? — Study Design & Path to Publication

Aligning the Mayo Clinic Datathon prototype with a publishable study

MIT–Mayo Clinic Florida Datathon spin-off · MIMIC-IV · Draft v0.2 (repo-aligned) · July 2026

Repo: github.com/tinaboya/Signal2Scan

How to read this document. This is the technical lead's working plan, reconciled against what the team has already built in the repo. Each section is tagged: **[BUILT]** = done in the datathon and kept as-is; **[TO ADD]** = needed to move from prototype to paper; **[DECISION]** = needs the physician lead before we lock it. This is a living Google Doc — comment inline.

Contents

0. Where the project stands

The datathon produced a complete end-to-end pipeline: a labeled CT cohort, a 4-hour pre-scan feature set, five trained classifiers with leakage-safe splits, and SHAP interpretation. In other words, the **experiment already exists**. The remaining work is not to design a study from scratch — it is to add the **rigor and validation** a datathon skips, and to write it up. This document therefore doubles as a study protocol and a gap list.

Core question (from the repo): can routinely collected ICU data, available *before* a head CT is performed, predict whether that scan will show an acute, clinically significant finding? Framed as scan-yield / appropriateness rather than a diagnostic risk score — a framing that is both clinically honest and easier to defend, because it is explicitly about ordered scans.

1. Background & Rationale

[BUILT — motivation already in README]

Head CT is ordered frequently in the ICU, but a large fraction returns no acute findings, exposing unstable patients to avoidable radiation and transport risk. Existing head-CT decision rules (Rothrock/Orlando, Covino, Scott-King) were almost all derived in the **emergency department** and achieve high sensitivity but poor specificity (the pooled Orlando meta-analysis: ~98% sensitivity vs. ~18% specificity). ICU evidence is far thinner (Chokshi 2016; Khan 2014; Purmer 2012; Balachandran 2009), and ICU patients break ED assumptions: sedation and ventilation depress GCS independently of intracranial pathology, AMS is near-universal, and baseline exams are often unobtainable.

Positioning for the paper: Signal2Scan targets the ICU gap directly, using routinely captured structured data (not a bespoke bedside checklist) to estimate scan yield. The novelty is the ICU setting + the pre-scan structured-data-only feature design + interpretable modeling.

2. Objectives & Hypotheses

[TO ADD — formalize before writing]

2.1 Primary objective

To predict whether a head CT ordered in the ICU will show an acute, clinically significant finding (Positive vs. Negative), using structured features from the 4-hour window preceding the scan, and to identify which routine signals drive that prediction.

2.2 Secondary objectives

- Quantify head-CT yield in the ICU cohort overall and by subgroup (ICU type, shift, indication signals).
- Compare an interpretable model (logistic regression) against flexible learners (RF, GBM, SVM, KNN) to test whether complexity yields clinically meaningful gains.
- Characterize the predictive signal with SHAP and reconcile it with clinical plausibility.

2.3 Working hypotheses

- H1. A small set of routine pre-scan signals (neuro exam deltas, coagulation status, indication context) carries most of the discriminative signal.
- H2. Flexible models improve discrimination only modestly over logistic regression, supporting a simple, deployable tool.
- H3. Predicted yield varies substantially across ICU subgroups, implying context-specific ordering behavior.

3. Data Source & Cohort

[BUILT] cohort construction & exclusions are implemented.

Source: MIMIC-IV critical care database + MIMIC-IV-Note for free-text radiology reports (linked via subject_id / hadm_id). Access is credentialed (CITI training + PhysioNet DUA); raw patient CSVs are non-redistributable, so only synthetic/example files live in the repo.

Head-CT identification: reports for CT HEAD W/O CONTRAST, CT HEAD W/ & W/O CONTRAST, CT HEAD W/ CONTRAST pulled from the note tables.

Key exclusion (a real strength): patients whose admission diagnosis is itself a neurological indication for CT (trauma, stroke, tumor, CNS infection, hydrocephalus, post-neurosurgery, seizure, anoxic injury, ...) are excluded, so the model does not simply learn 'structural neurological disease → imaging.' ICD-9/10 lists are in ICD_codes to_exclude.docx.

[DECISION] Confirm the exclusion list is clinically complete and that the neuro-indication exclusion is the intended target population for the paper's claim (i.e., 'ICU patients scanned without a pre-existing structural neuro indication'). This defines exactly who the tool would apply to.

3.1 Cohort composition (from the repo)

CT label class	Count	Handling in modeling
Negative (no acute finding)	3,383	Modeled (majority class)
Positive (acute finding)	1,004	Modeled (minority class)
Unclear	169	[DECISION] — excluded? sensitivity analysis?
Post-surgical / Unchanged	82	[DECISION] — excluded? separate class?

[DECISION] Pre-specify, once, how Unclear (169) and Post-surgical/Unchanged (82) are handled, and state it in the methods. Current modeling is Positive-vs-Negative; make the exclusion of the other two explicit and justified rather than incidental.

4. Outcome Labeling

[BUILT] + [TO ADD: validation — the #1 gap]

Labels are assigned by rule-based regular expressions over the IMPRESSION / FINDINGS sections into Negative / Positive / Post-surgical-Unchanged / Unclear, with Positive sub-typed into Hemorrhage, Infarct, Fracture, Mass Effect, Hydrocephalus, and with negation handling. Implemented in `Classifying CT reports.sql`.

The gap that reviewers will catch immediately: the regex labels have no reported validation against a clinician-adjudicated ground truth. Without it, the entire outcome — and every downstream result — is questionable. This is the single highest-value addition to the project.

4.1 Label-validation plan [TO ADD]

1. **Gold set:** draw a stratified random sample of ~300–500 reports (over-sampling Positive and Unclear).
2. **Double annotation:** two clinicians label against a written codebook; a third resolves disagreements. Report Cohen's / Fleiss' κ .
3. **Pipeline metrics:** compute the regex labeler's sensitivity, specificity, PPV, NPV, and confusion matrix vs. the gold set; report per-subtype too.
4. **Acceptance gate:** agree a threshold (e.g., $\kappa \geq 0.80$) below which the labeler is revised before any modeling result is considered final.

[DECISION] Physician lead to (a) sign off the positive-finding codebook and borderline handling (small stable subdural, microhemorrhage, chronic-only), and (b) set the κ acceptance threshold.

5. Features (4-hour pre-scan window)

[BUILT] candidate predictors + MIMIC itemids in `predictive_variables.docx`.

Domain	Variables
Neurological	GCS (eye/verbal/motor/total + delta from baseline), RASS sedation, pupil size & reactivity, CAM-ICU delirium, seizure flags
Vitals	MAP, systolic BP, heart rate, SpO2, respiratory rate, temperature
Labs	INR, PTT, platelets, sodium, glucose (+ coagulopathy / hyponatremia flags)
Medications	New vasopressor, new antiepileptic
Context	ICU type, shift, hours since ICU admission, demographics (age, sex, race)

Design strength: features use only data available before the scan and explicitly encode neuro deltas (GCS change), which partly addresses the sedation-confounds-GCS problem via RASS + baseline-referenced GCS. Worth stating this as a deliberate methodological choice in the paper.

[TO ADD] Report and justify missing-data handling. The repo drops high-missingness columns (pupillometry, temperature, PCO2, MAP); make the drop rule explicit and consider a sensitivity analysis with imputation, since dropping may discard clinically important signal (pupils).

6. Modeling

[BUILT] Positive-vs-Negative, 5 models, leakage-safe splits, SHAP.

- **Task:** binary Positive vs. Negative.
- **Models:** Logistic Regression, Random Forest, Gradient Boosting, SVM, KNN.
- **Splits:** `GroupShuffleSplit` on patient ID (no patient in both train and test — correct leakage control).
- **Interpretation:** SHAP feature importance. Implemented in `Final_dataset` and `code/ML_models_CT_head.ipynb`.

[TO ADD] For publication: (a) report a clear primary model and pre-registered primary metric rather than a model zoo; (b) add class-imbalance handling (class weights / resampling) and report AUPRC prominently; (c) nested or repeated CV with confidence intervals, not a single split.

7. Evaluation

[BUILT: AUROC/AUPRC/accuracy/Brier] + [TO ADD: calibration plots, decision curves]

Metric	Status	Why it matters
AUROC	Built	Overall discrimination
AUPRC	Built	Robustness under imbalance (Positive is minority)
Brier score	Built	Calibration (numeric)
Calibration curve	To add	Reviewers expect the plot, not just Brier
Sensitivity / NPV @ threshold	To add	Clinical crux — safe rule-out of low-yield scans
Decision curve / net benefit	To add	Does it actually help ordering decisions?

7.1 Subgroup & sensitivity analyses [TO ADD]

- Performance by ICU type and shift (already collected as features).
- Sensitivity to Unclear/Post-surgical handling; alternative positive-finding definitions; imputation vs. column-drop.
- Fairness check across available demographic groups (age, sex, race).

8. Interpreting the Current Findings

[DECISION — needs clinical explanation before publication]

The datathon models surfaced these directions:

- Higher PTT, INR, and low platelets → more likely Negative CT.
- Lower GCS motor & verbal → more likely Positive CT.
- Higher respiratory rate → more likely Positive CT.

The coagulation direction is clinically counterintuitive — one might expect coagulopathy to raise bleeding risk and thus Positive scans. The observed inversion is most likely a **selection / collider effect**: coagulopathic patients may be scanned more liberally (lower threshold), diluting the Positive rate, or coagulopathy may co-occur with sedation/metabolic causes of AMS that scan Negative. This needs an explicit clinical interpretation (and possibly a stratified analysis) before it goes in a paper, or it becomes a reviewer flag. The lower-GCS→Positive direction, by contrast, is clinically expected and reassuring.

9. Datathon → Publication: Gap Summary

Priority	Gap	Action
1 (blocking)	No label validation	Clinician gold set + K vs. regex labeler (Section 4.1)
2	Counterintuitive coag finding unexplained	Clinical interpretation + stratified analysis (Section 8)
3	No calibration plot / decision curve	Add both; frame around ordering decisions (Section 7)
4	Unclear/Post-surgical handling implicit	Pre-specify and justify (Section 3.1)

Priority	Gap	Action
5	Missing-data handling implicit	State drop rule; imputation sensitivity analysis (Section 5)
6	Model zoo, single split	Primary model + CV with CIs + imbalance handling (Section 6)
7	Selection bias not framed	State scope: 'yield among scanned ICU patients' (Section 1/2)

10. Workstreams & Task Allocation

Maps to columns in `Datathon-DataTracking.xlsx` (already in the repo). Owners/dates set at kickoff.

#	Workstream	Status	Owner
A	Label validation (gold set + κ)	TO ADD — priority 1	Technical + physician leads
B	Clinical interpretation of findings	TO ADD — priority 2	Physician lead
C	Calibration + decision-curve analysis	TO ADD	Technical lead
D	Cohort edge-case handling	TO ADD	Technical lead
E	Modeling hardening (CV, imbalance)	TO ADD	Contributors
F	Manuscript draft	Not started	All
G	Cohort / labeling / features	BUILT — maintain	Technical lead

11. Publication Venue — options to decide

[DECISION — not yet chosen]

The framing (interpretable, clinically-motivated, utilization-focused) fits several venues. Trade-offs:

Venue type	Fit	Consideration
Med-informatics journal (e.g. JAMIA, npj Digital Medicine)	Strong	Values clinical validation + interpretability; label-validation gap must be closed first
Clinical / critical-care journal	Strong if yield-focused	Emphasize the clinical decision (imaging utilization), less on ML machinery
ML-for-health venue (ML4H / CHIL)	Moderate	Faster; expects methodological novelty — the ICU pre-scan design + interpretability angle

Recommendation to discuss: target a med-informatics or critical-care journal, because the contribution is clinical utility and interpretability rather than a new algorithm — but only after the label-validation gap (priority 1) is closed, since that is exactly what these venues scrutinize.

12. Open Decisions for the Next Meeting

1. Confirm the neuro-indication exclusion defines the intended target population (Section 3).
2. Finalize the positive-finding codebook + κ acceptance threshold (Section 4.1).
3. Decide Unclear (169) and Post-surgical (82) handling (Section 3.1).
4. Provide a clinical read on the counterintuitive coagulation finding (Section 8).
5. Agree missing-data strategy (drop vs. impute) (Section 5).
6. Choose publication venue and authorship/contributorship policy (Section 11).